

Essential Design Concepts and Principles to Include in Maritime Domain Awareness Platforms

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Introduction / Background

Maritime Domain Awareness (MDA) relates to being aware of activities occurring in the marine domain that may have an impact on the security, safety, economy, and environment of the ocean (Tetreault, 2005). The goal of MDA is to develop understanding and transparency of activities and threats at sea (Bueger, 2015). This includes the people, infrastructure, cargo, vessels, and any other forms of transport on the waters (Transport Canada, n.d.). Thus, it must collect sufficient and accurate information of ongoing maritime activities to identify potential threats, make informed decisions, take appropriate action, share information with relevant partners, and respond to maritime activities by working with the government incorporating situational awareness to minimize resulting damages (Transport Canada, n.d.). However, MDA is challenging as collecting and interpreting the data to develop systems to detect such anomalies is socially, politically, and legally complex with different laws, rules, and practices governing different areas (Bueger, 2015).

A common way to collect information like vessel detection, movement, classification, identification, and consistent monitoring of the vessel which supports surveying the waters is with Automatic Identification System (AIS) (Tetreault, 2005). AIS transmits a vessel's position, showing its position to other vessels and organizations. The International Maritime Organization and other management entities require large ships like commercial fishing vessels to use AIS to locate their position as it prevents collisions (Global Fishing Watch, n.d.). More than 400,000 AIS devices have been used for vessel location, identity, course, and speed information annually (Global Fishing Watch, n.d.). This information is critical for MDA, and is being used in existing platforms and interfaces like Global Fishing Watch (GFW) and Skylight (SK) which are commonly used for risk assessment. However, these platforms depend largely on and are designed around AIS data, meaning the quality of information collected is essential for it to be productive (Emmens et al., 2021). However, a huge limitation is that only two percent of the world's fleet use AIS, meaning many vessels are not tracked (Global Fishing Watch, n.d.). Thus, situational awareness cases cannot be made due to lack of understanding of interaction between small-scale fleets (Belhabib, personal communication, 2022).

Also, AIS does not provide concrete data. Although these platforms are risk assessment tools, it requires human analysis to interpret the AIS data. This is a limitation on the functional requirement which is to provide awareness of maritime activities and identify possible threats, taking appropriate actions as stated previously since these platforms are unable to inform users of these events. Additionally, AIS itself can be unreliable and insufficient for users' needs, making the human analysis more difficult and subjective. It is common for vessels to turn off their AIS signal to hide their whereabouts when engaging in illegal

activity (Emmens et al., 2021). It is also normal and legal for vessels to turn off their AIS signal when they feel unsafe in some region (Kolla, personal communication, 2022). Another weakness with AIS data is that it can also be “noisy” due to the equipment used, external factors, humans, and dense traffic, making the data unclear (Emmens et al., 2021). This is why Nautical Crime Investigation Services (NCIS) is developing their own products to try and minimize these limitations. These products are *Edith*, a tracker to provide more coverage of the global fleets, *Hava*, a database filled with past crimes, and *Grace*, an AI powered risk assessment tool that uses *Edith*, *Hava*, and machine learning to predict and identify suspicious activities and notify the appropriate organizations to intercept before it occurs and improve maritime security (Belhabib, personal communication, 2022). With machine learning, *Grace* should be able to predict with high accuracy and certainty what and where suspicious activities occur and who is involved. Then, it bridges the gap between the information collected and the information interpreted, eliminating the dependency for human analysis (Ghattan, personal communication, 2022).

For this report, we will focus on the design implementation of the MDA platform, *Grace*. Besides from the information collected by MDA platforms, they must be able to convey the information so users can interpret the necessary information for their needs. User experience (UX) and user interface (UI) affects how users learn to use and interpret the system as it is the link between the two entities (Pratama, 2020). UX is based on research and strategy, dealing with the *overall customer experience* with the interface from end-to-end (Masters in Data Science, 2020). The goal here is to understand the user and their needs, goals, and struggles in order to build a product that leaves a positive experience for the customer (Masters in Data Science, 2020). UI then uses the UX research, organization’s branding, and design concepts to optimize the interaction between the users and the interface to implement this positive experience for the user. It relies on the users’ needs to understand what specific components help them achieve their goal, thus, it is more focused on *interaction* between the user and the interface (Masters in Data Science, 2020). Thus, for an interface to be successful, it must consider the users during the design process (Guntupalli, 2008). Not only does this minimize human error, but it also helps design decisions match the users’ expectations, allowing them to use the systems successfully (Guntupalli, 2008).

Then, besides the functional requirement of MDA platforms, the purpose of this study is to identify other design requirements that *Grace* should address, recommend which design concepts and principles are essential to include for an MDA platform, and provide examples of potential features that can support these requirements. This research uses a competitive review, interviews, and usability test.

Research Scope

As *Grace* is a new product that is currently being designed, our research goal is to understand how existing MDA platforms are currently supporting their needs and how their user experience is like with these interfaces. The research questions we hope to answer are (1) how helpful are the features / which features are helpful; and (2) is what is displayed on the interface enough for effective and productive interpretation. This will help us learn what features are necessary to include in *Grace*, provide insight on how we can improve user experience, and eliminate assumptions or misunderstanding in users' context, needs, expectations, and abilities which can then minimize human error (Sharp et al., 2019).

Our research involves conducting research with end users to identify their habits, gaining a better understanding as to how we can improve user experience and usability of MDA platforms. This pre-design stage will generate the requirements needed in *Grace*, leading us to determine the design concepts and principles for the initial design phase in the early design stage. Though this report will not cover all of the early design stage, it will contain recommendations of potential features with some lo-fi prototype examples to consider in *Grace*.

Related Company Goals

Who is NCIS?

Nautical Crime Investigation Services (NCIS) is a tech corporation that uses a human-centered approach to provide responsible and ethical solutions to detect crimes at sea and bring maritime domain awareness. They provide tech solutions to survey waters and strive to be culturally aware, ethical, and scientific. NCIS wants their products to be accessible to their clients' needs, regardless of their responsibilities and backgrounds. This means they want their product to be usable by the government, maritime insurers, seafood suppliers, journalists, researchers, NGOs, citizens, fishermen, and many other groups.

Who are the customers?

The government, including sectors like the monitoring, control, and surveillance (MCS) and enforcement agencies want to ensure fishing vessels are following regulations and pursue those that are participating in illegal activities (Whyte, 2016). Fisheries officers are responsible for enforcing the federal and provincial regulations for protecting fish, wildlife, and natural resources (WorkBC, 2022). So, they will want to use *Grace* to monitor and detect whether vessels enter a prohibited zone, compromising ocean safety. MCS workers are focused on deterring illegal, unreported, and unregulated (IUU) activities relating to the sea (Government of Canada, n.d.). Thus, they may be interested in seeing suspicious activity occurring at sea. Maritime insurers are interested in the conditions of vessels and the safety of the people on the vessels, as they are liable for bodily injuries or deaths of a person, and the damage of the vessels (government of

BC). Then, they may use *Grace* to monitor encounters, collisions, or potential dangers to people like piracy, human rights violations, and more. Seafood suppliers are interested in how their product is sourced, so they may use *Grace* to follow their suppliers to understand where and how their seafood products are being caught (Whyte, 2016). Maritime journalists and researchers have the responsibility to investigate and report news regarding the ocean, including ocean health, impacts of fishing and other ocean activities, and suspicious vessels and activities. So, they may be interested in a variety of features offered by *Grace*. NGOs and citizens may also be interested in a variety of topics, including seafood sustainability and suspicious activities at sea, and may make it their responsibility to report these findings to the respective authorities (Whyte, 2016). Finally, fishermen could use *Grace* so other vessels are aware of their location for transshipment and encounter purposes, or to prevent collisions. Therefore, *Grace* should provide sufficient and effective features for each of these groups and their specific tasks to satisfy their functional requirement.

Geographic

Also, NCIS wants to develop their product to be as inclusive and accessible as possible, regardless of where the users are located in the world. Global Fishing Watch and Skylight are MDA platforms that are very successful in North America, however, *Grace* aims to be successful in underdeveloped countries, like the subregion of Africa. The interface design depends largely on cultural differences, as different cultures will have different challenges and opportunities, affecting how people perceive and interact with each other and other entities, like computer systems (Halskov & Christensen, 2018). Then, *Grace* must be designed to incorporate user behaviours and values from these areas and prioritize creating relatively universal features that are applicable to all end users.

Thus, the design requirements for *Grace* should be based on these users' expectations, capabilities and fallibilities, while maintaining its functional requirements and providing a good user experience.

Approach & Recruitment of Participants

Overview

First, we chose MDA platforms to investigate and explore. We chose Global Fishing Watch (GFW) and Skylight (SK) as they are similar risk assessment tools that show visualizations of vessel-based activity at sea. We used a triangulation approach to enhance validity, collect more in-depth data, and interpret the problem in different ways (Nightingale, 2020). Triangulation involves collecting and analyzing data from more than one source, resulting in researching the same problem from multiple perspectives (Nightingale, 2009). Here, four methods were used, including a short interview with the internal stakeholders of NCIS,

an independent competitive review, and a usability test in the form of a direct field observation followed by a short interview with external stakeholders.

Interview (Internal Stakeholders)

The purpose of this initial interview was to understand the goals and vision for the company and their products (Appendix D.2). The participants were the two co-founders of Nautical Crime Investigation Services (NCIS). The interview was open-ended to allow probing to understand deeper, but questions and topics were used as a guide. It covered questions regarding them as a company, what they think a successful MDA platform should look like, and their expectations for their products, specifically *Grace*. It also involved their opinions on GFW and SK, such as the purpose for using these interfaces, how they support or not support their work, any problems they may run into, what features are specifically helpful, and how easy they find these platforms to be. Thus, most of the data collected was subjective and qualitative, as much of their answers were from their perspective. This data was then used to determine the human need and central tasks for intended users.

The human need is to bring maritime domain awareness and detect and deter crimes.

The central tasks are to detect suspicious activity accurately and efficiently.

Competitive Review (Independent)

A competitive review is a form of evaluation of the interface where an individual reviews designs based on their usability experience and knowledge (Schade, 2013). The individual who reviewed it was the lead researcher in this study. This was to familiarize myself with the platforms and see what works and what does not from my perspective (Appendix D.3). It involved exploring GFW's map directly, using task examples I created to see if the interface was able to support the task. As for SK, video tutorials were used to understand the workings of the interface and what features it offered. During this stage of research, strengths and weaknesses of both interfaces were noted down and then categorized to UX/UI specific qualities versus algorithmic qualities, as this research will only focus on the UX/UI aspects. This data was also mostly subjective and qualitative as it was from the reviewer's perspective. This data was then used to create evaluation goals for the observation and short interview.

The following are the evaluation goals:

- 1) *How helpful is the map / interface for finding vessels?*
- 2) *How helpful are the filters in supporting their tasks?*
- 3) *Is the available information displayed on the interface sufficient for effective interpretation?*

Usability Test: Direct Field Observation + Interview (External Stakeholders)

A usability test is another way to evaluate an interface. It helps determine whether an interface is usable by the intended user population to carry out tasks for which it was designed for (Sharp et al., 2019). We used this method to identify where we could improve MDA platforms in terms of accuracy, performance, and user satisfaction.

This usability test was structured as an observation and follow-up interview. An observation was used to understand the context of when and how end users would use MDA platforms like GFW and SK. It allowed us to watch the user interact with the interface directly. This observation session was paired with a short follow-up interview to dig deeper and summarize their overall experience of using MDA platforms. From the initial interview with the internal stakeholders, we developed an understanding of who the end-users are for *Grace*.

The main users will be those from developing regions, especially around Africa, who survey the waters. This reflects in NCIS' branding strategy, wanting to be as culturally aware and inclusive as possible, and making their product accessible globally. Thus, the recruitment strategy was to first use convenience sampling, reaching out to referrals from the internal stakeholder, then using LinkedIn and online platforms to find those working in the maritime criminality industries internationally, including Africa, UK, and Asia, and then following up with past interviewees asking if they are able to refer me to others working in the industry internationally. Emails and short messages via LinkedIn and WhatsApp were sent out detailing the purpose and structure of the study to potential users. Four observation sessions were set up, three using GFW and one using SK.

This session was a virtual Zoom session. It included completing four tasks, either on GFW or SK, which aimed to answer the evaluation goals stated previously. These four tasks were the same across users and sessions, regardless of interface:

- 1) *Can you find Long Wang Chyun NO37 in 2021 and tell me a bit about what you know about the vessel from the interface?*
- 2) *Can you find a Spanish trawler fishing in a prohibited zone?*
- 3) *Can you identify an encounter between 2 vessels and tell me what each of their flags are?*
- 4) *Beside fishing related activities, can you identify any other crimes from this interface?*

A coding sheet was used to guide data collection during the observation and the data was qualitative, quantitative, subjective, and objective. Some examples of types of data were the time it took to complete a

task, whether the task was successful, what features were used, body language and facial expression, and any comments made throughout the observation.

The follow-up interview was conducted immediately after the observation and consisted of 9 questions. Most were open-ended, allowing users to elaborate on their answers, and many were also paired with a Likert scale to allow for better comparison between participants. The data collected was qualitative, quantitative, subjective, and objective as well. Some of the questions covered included what they did for work, how familiar they were with the MDA platform they used, how easy was the platform, and other examples of tasks they would do for their work.

These findings, along with the abundance of data from the competitive review and initial interview, were analyzed and summarized to generate requirements for *Grace* (Appendix D.1).

Key Findings & Requirements

Global Fishing Watch Qualitative Analysis

From our affinity diagram for GFW observations, we identified themes relating to three strengths, three weaknesses, and one limitation (Appendix C1.1).

Strength: Signifiers help with interpretation, learnability, and usability

Something done really well by GFW is their use of signifiers to communicate how to use the interface to a new user. GFW uses colour identification and legends to indicate what feature is at work on the map which users found helpful as they referred to it during their tasks. Another helpful feature is the information button *i* next to each feature that describes what the purpose of the feature is and how to use them. We saw most users using it when they were looking for filters / features to help them with their tasks, especially in task 4, suggesting it is helpful for discovery purposes. Finally, one of the most helpful features we saw from the observation was that every user took advantage of the question mark bubble (Appendix A.4). This bubble stood out from the map, encouraging users to click it to reveal information on how to navigate the interface, improving learnability.

Strength: Encourages discoverability

Another key pattern observed among these users was that most felt comfortable playing around with the interface. When we asked users to find other criminal activities unrelated to fishing, everyone went through every visible button and icon on the screen, including the filter panel on the left side, the buttons on the right, the timeline on the bottom, and even the toolbar furthest to the left (Appendix A.1). This led

to users discovering the other offerings of GFW like the type of sources they get their information from (AIS, VMS, SAR). Users also felt very comfortable clicking on the grids on the map, and any buttons that popped up, like the question mark bubble as mentioned earlier.

Strength: Helpful features offered

Every user used the *Search for vessels* function and also played around with the timeline. Users also commented about how many helpful and relevant features GFW offered, including the different reference layers and source types. Every user mentioned GFW had at least one feature that would be helpful in their line of work, whether it was to help them monitor vessels, deter illegal fishing, detect entry / jurisdiction violations, or other responsibilities. However, this may have traded off the efficiency of the platform as users also commented about it being slower to load and refresh, leading us to our first weakness.

Weakness: Slow and heavy / overwhelming interface

A consistent finding across each GFW observation was that the interface was very slow. Although GFW has attempted to make as many of their features as visible as possible, it may be the reason as to why the system takes a long time to refresh and reload updated queries. Every user tried refreshing assuming it would make the page load faster. One user turned their hotspot on as they thought it was their network. Fortunately, most of the users found that by shortening the timeline, GFW could load slightly faster, suggesting the amount of data probably contributed to the efficiency. This shows that some users understand the system to be slow incorrectly, affecting their learnability and experience with the interface.

Weakness: Unclear signifiers and poor feedback results in misinterpretation

There were a few cases when users' expectations of their actions did not match the system status. For one, every user who used GFW had difficulty figuring out how to filter for flag country and gear type. From the observations, it was because users had difficulty finding the filters feature, suggesting this signifier was poorly designed. Guidance was given to all four of the users to find the filter icon and the task was easy to complete after locating where the feature / icon is on the interface. Second, users assumed the text under the list of vessels reading *+45 more* is clickable (Appendix A.2). This may be because their mental models had led them to believe the text would show what the other 45 vessels were. From previous experiences, users may have seen similar texts located at the bottom of a list reading *Load more*, *See more*, or *See 45 more*. This is a common signifier used to show more of similar items. Thus, using this text under a list may have been the cause of poor feedback as most users who used GFW performed the same action with similar expectations. Similarly, some users tried clicking on the circle that shows up with a number when looking for encounters (Appendix A.3). GFW uses these circles to indicate the number of

activities occurring in the area, but users incorrectly clicked on these circles thinking it would do something. This error suggests users are not learning the interface effectively.

Weakness: Confusion around using some of the features

Although users found the features helpful, most users found it difficult to learn and use initially. There were many comments and questions “How do I do X?” and we saw a lot of hovering over features as if lost or confused. This reflected in the interview as most users mentioned there was an initial learning curve, especially around interpreting the information with comments like “Is this what I should do?” and “I don’t know what X means”. There was also one user who requested a user manual several times, suggesting the interface was not as intuitive as it could be. Furthermore, there was particular difficulty for task 2, we hypothesize it was due to the extra step of human analysis and interpretation, though this critical incident will need to be looked into further.

Limitation: More concrete information requested

From the follow-up interviews, almost every user had mentioned they wished there was a feature that could inform them of more specific activities. The types of activities mentioned included understanding what activity was occurring between two vessels during an encounter, what type of vessels are shipping where, which areas are prohibited or authorized for specific types of vessels, and more. These activities would help the MDA platform fulfill its functional requirements of providing maritime awareness and detecting and deterring suspicious activities occurring at sea.

Skylight Qualitative Analysis

Only one user used this platform due to users needing an account to access this platform. Since only one user was observed with this platform, patterns were not observed, but key differences in visual design and how the system works between SK and GFW were identified.

Visual hierarchy

The biggest difference between SK and GFW is the visual hierarchy which is the structure of information on a page that communicates order, emphasis, and relationship. As seen in Appendix B.1, the features are located in the toolbar at the top of the interface instead of a side panel. SK uses dropdowns as its main way of encompassing features they offer, while GFW uses a scrollable side panel. A strength that comes with using dropdowns is it declutters the interface. However, it needs to ensure the names for each dropdown option is an accurate reflection as to what is in the hidden menu since these features will be invisible to the user until they click on these tabs. Contrastingly, GFW’s side panel may offer more

visibility. Yet, it has a very skinny scrollbar which users may miss and result in them thinking there are not any more features offered than what they see at the point of entry. This conclusion resulted from one observation where the user was not aware there were more features offered past what they were able to see without scrolling. Furthermore, using a side panel to show every feature offered also means using different fonts and visual designs to group sections and separate ideas. For instance, for one feature, there are different sized fonts, type of texts, colours, input types, labels, and imagery / icons for different ideas and groups (Appendix A.1). These designs must be applied to every feature offered to provide consistency and also be well understood so users can learn and use it easily.

Visual cues

Another difference we noticed was the type of visual cues used between the two MDA platforms. Visual cues are the elements our eyes process during visual perception and are virtually everything on the interface that we can see (Tidwell, 2010). One example of this is how SK uses text as its primary or complementary way of signifying possible actions. For instance, *Filters* is complemented with a standard filter icon to indicate it is the button to filter (Appendix B.1). SK also includes the number next to each feature to indicate the number of options selected. Another example is how SK uses a map that labels every country unlike GFW's interface which only uses the silhouette of the continents. A third example is the way the platform translates the location of a vessel. While GFW uses grids, SK uses specific icons to indicate the type of activity and where in the ocean it occurred. But, this is dependent on how they want to show information. Since GFW attempts to show every single occurrence of when a vessel is fishing where, icons may cause the interface to be too cluttered. Thus, with grids, GFW provides a more seamless look and also uses different shades of colour to indicate where more or less fishing occurs. Contrastingly, SK can use icons as it only shows the last point where the vessel was detected for some activity.

How it works

SK and GFW also work very differently. Upon loading up the system, SK waits for users' queries before loading anything on the interface, while GFW loads immediately with almost every vessel collected from AIS and VMS as this is their default status. Furthermore, SK organizes vessel data and activity differently from GFW as it has a separate page listing every activity the vessel has been involved in. However, unlike GFW, SK only shows the most recent position of the vessel and does not show historical tracks. And, a smaller duration is tracked for each event that has occurred. Thus, this characteristic may be a reason as to how the data loads a lot faster than GFW as there is less data upon updating. Another difference is the type of features and filters offered. Although SK and GFW both offer basic features that apply to most

users, including finding and monitoring vessels and identifying boundaries and jurisdictions, SK also covers activities not related to fishing, including speed and entry activities.

Quantitative Results & Analysis

We found the average of the following metrics for MDA platforms from every user, regardless of which platform they were using (Figure 1).

Metric	Out of 5
Familiarity (1 = not familiar, 5 = familiar)	2.75
Navigation difficulty (1 = easy, 5 = hard)	2
Interpretation difficulty (1 = easy, 5 = hard)	2.44

Figure 1: Average of ratings from GFW and SK users (Personal Experience and Opinions)

These were subjective ratings and not every answer matched their actual performance. Thus, we also looked at the time it took for them to complete the three tasks (Figure 2).

	GFW Average Time (s)	SK Average Time (s)
Task 1	87	10
Task 2	440	120
Task 3	340	120

Figure 2: Average time to complete tasks 1, 2, 3 on GFW and SK.

We understand the time it took to complete the task does not necessarily reflect efficiency without accuracy, thus, we also analyzed their behaviours and general accuracy for each task to support the findings on efficiency and their subjective ratings.

GFW Accuracy Analysis

Task 1: Every user except one found the vessel on the map and its tracks. Every user correctly interpreted the information regarding the vessel, including their flags, MMSI, call sign, gear type, but some uncertainty and confusion arose regarding the type of fishing and what the dates mean on the interface.

Task 2: This task was one of the harder tasks as it required experience and knowledge of what a prohibited zone was. Most users assumed where a prohibited zone was based on their experience and many used EEZ and MPA reference layers. Also, most users initially tried to use the search vessel

detection feature which they had previously used for the first task. Specific descriptions were typed in this search bar, including different variations and combinations of “Spanish”, “trawlers”, and “ESP”. One user managed to find a Spanish trawler, but no effort was made to identify where the vessel was on the map to check if it was fishing in a prohibited zone. The other users had trouble finding features to help them with this task, thus, guidance was given towards where the filter feature was.

Task 3: The difficulty was around locating the necessary features to complete this task, however, once the *Encounter* feature was found, it was relatively easy to identify an encounter and the flags of the vessels involved. Only one user was not able to identify the flags of the encounters.

SK Significant Findings and Analysis

Note that only one user was observed. This user completed the three tasks accurately and was very familiar with SK, thus, this data may not be an accurate representation of the end users. Still, the difference in time the user completed these tasks compared to users using GFW may indicate some design properties that may be more helpful to include in MDA platforms. All three of the tasks were completed significantly faster than users who used GFW.

Task 1: This difference may be because the search function is very visible at the top of the interface with the label *Search for vessel* clearly identified. The information interpreted was also different, as this user focused more on the events the vessel was involved in, including where it fished, duration it fished, and where it was last found. Interestingly, SK did not have as many vessel characteristics found for the same vessel used in GFW, showing an unknown call sign, IMO, and vessel type.

Task 2: This difference may be because of how SK organizes their filters. Unlike GFW, SK uses a different filter icon with the label *Filters* next to it, making it more visible. Thus, this may be easier to locate than the filter icon on GFW. Furthermore, SK allows users to draw and add zones onto their map. This user already had a few zones drawn and had some knowledge as to which areas may be prohibited, making finding a Spanish trawler in a prohibited zone a lot easier as she just had to see if any vessel crossed into the areas.

Task 3: This difference may again be due to the visibility of the feature. Though users who have never used SK before may spend longer finding this feature, SK only has three dropdowns to choose from, making it a lot faster to go through to find the feature the user is looking for. Also, users will have to know that an encounter between two vessels is called a *Standard Rendezvous* in SK, which may also be a

point of confusion without training as SK does not explain what it is on the interface. However, assuming users have this knowledge, they can easily filter the database to only include *Standard Rendezvous* and click on any of the icons that appear which will automatically show the vessels and their information in a pop-up. This makes identifying the flags more seamless instead of having to do several steps to determine the vessel information.

Conclusion

We conclude that the existing MDA platforms, specifically GFW and SK, have helpful features to support users in their tasks and responsibilities. Specifically, every user used some sort of vessel detection search function which proved helpful for finding vessels. For GFW, the average time for task 1 was around 1.5 minutes, and the user using SK completed it in 10 seconds. Similarly, task 3 was well supported with the *Encounter* or *Standard Rendezvous* feature. Users using GFW completed task 3 in less than 6 minutes with high accuracy, while the user using SK completed it in 2 minutes. This suggests features supporting tasks that do not require much subjective human analysis / interpretation works well in supporting users.

However, we found some areas for improvement in regards to some design choices for signifiers that play an important role in serving the functional requirement. Specifically, there was difficulty when locating certain features, such as the *Filters* feature. Since every user using GFW could not locate it without help, we conclude this signifier does not match their mental model, and thus, the interface lends itself to the possibility of confusion, hindering learnability and usability. We then conclude that GFW does not provide sufficient information for users to understand or interpret the signifiers to take advantage of the features offered. Then, the interface is not maximizing its output and thus, is not fulfilling its functional requirements. Moreover, although users mentioned the learning of the interface is relatively easy considering the type of information provided and the context these MDA platforms are used in, we observed a lot of misinterpretation and navigation issues as seen in the affinity diagram (Appendix C.1). Furthermore, GFW users had difficulty with task 2 when they had to interpret whether a vessel had fished in a prohibited zone. The average time here was 7 minutes, with multiple users reaching past 10 minutes and every user noted the assumptions they made. This suggests GFW provides insufficient information for effective and efficient interpretation when it comes to subjective matters as well.

Then, to improve the overall interface in terms of usability and user experience, we determined the requirements needed for MDA platforms, specifically *Grace*:

- 1) *Functional: detect / monitor vessels and predict suspicious activity with high certainty*
- 2) *Usability: easily learned and retained*

3) *User characteristics: globally understood*

As stated previously, detecting and monitoring vessels and predicting suspicious activity with high certainty is a necessity as they are the central tasks for *Grace* and virtually the ideal MDA platform, making this an important requirement. *Grace* should also be easy to learn and retain so users can understand and interpret the information accurately and effectively. Also, this requirement allows *Grace* to compete with established MDA platforms as many users will not trade off switching platforms if it is too difficult to use. Finally, *Grace* should be globally understood to help with interpretation and also improve findability and discoverability of the features offered. And, it gives *Grace* a competitive advantage as it becomes more accessible, attracting a wider range of users.

Recommended Solutions for General Design: Design Concepts and Lo-Fi Prototypes

Based on the requirements, we propose three design concepts to focus on for *Grace* with three lo-fi prototypes of specific features based on these concepts.

Design Concepts

An affordance is the relationship between the physical object and the user (Norman, 1988). *Grace* must afford to do what GFW and SK can do, like detecting and monitoring vessels, reporting vessel information, identifying activities like transshipments and encounters, and filtering through these options to make it easier for users to use and read the interface. But, *Grace*'s differentiating point is to be able to predict potential crimes and identify suspicious activity at high certainty. This serves their functional requirement of providing awareness of maritime activities and preventing crimes to close the gap between human analysis and risk assessment. And, with the filtering feature, it supports the usability requirement.

Signifiers are perceivable cues about the affordance (Norman, 1988). They should always be visible as they communicate the possible actions the user can perform, directing them to the affordances that are offered. As *Grace* is a system that offers complex information, there are features that support the users in doing their tasks. Though this can benefit a wider range of users, it can be overwhelming for individual users as they are usually looking to use only a couple features. Thus, signifiers are useful to help users understand what actions they are able to carry out without having to do too much mindless exploring on their own, as there may be too many features for them to try out for a single task. These signifiers must also be understood by users of different backgrounds and cultures to reflect the end-users of *Grace*, meaning they must be universally transferable.

Feedback communicates the results of the action taken so the user understands the consequences of their action. It should be informative, immediate, planned, and prioritized to help the user learn and use the system, and it should include how the user uses this feedback in the future (i.e. doesn't make the same mistake or understands what they did). It can take the form of visual, tactile, audio, or more, as long as users are notified of some change. The goal is to ensure the feedback output can communicate the result effectively so users are not left guessing the consequences of their actions. *Grace* must provide “good” feedback for every feature they offer. Good feedback means the system is responding to an action that the user can correctly interpret to either continue with their workflow or discover an error and backtrack to fix their mistake. The feedback should help them learn and work through the system their first couple times, while solidifying their knowledge of the possible actions of the system for future use.

Lo-Fi Prototypes

We used GFW's interface and designed some low fidelity prototypes for specific features to see these concepts at play.

Colour Code Activities

Again, **affordances** are the possible actions while anti-affordances are impossible actions (Norman, 1988). Affordances can be perceivable, but some are invisible, making it difficult to know how the object can be used and what actions are possible. It is the designer's responsibility to make these invisible affordances visible to the user. Thus, affordances and anti-affordances must be discoverable and understandable so users can perceive them with their senses (Norman, 1988). This can be done by making **signifiers** very visible to draw users' attention towards it. One way to execute this is to implement focal points which are deliberate cues to draw viewers' attention and are used to point out areas of interest (Rusonis, 2022). This is useful when developing novel features as users will not have previous experience to guide them in finding and using these features. Then, it is important to implement **good feedback** as users need to understand the consequences of their actions to learn such affordances.

For example, we can label specific activities at sea using different colours (Figure 3). Since one of the affordances is to detect suspicious activity at sea, including transshipment, illegal fishing, drug trafficking, illegal entry, and more, the dots can be coloured a specific bright colour if vessels are predicted to be involved in any of these activities. A transshipment violation can be coloured green, illegal fishing can be coloured purple, drug trafficking can be coloured red, and illegal entry can be coloured yellow. If any vessel is involved in more than one, they can be coloured orange. Though these colours are chosen arbitrarily, this works because these bright coloured dots will stand out in a sea of

black dots, leading users' to try and uncover what they mean. They can eventually click on these dots, and *Grace* can show a pop-up or a page that lists the information of the vessel and the activities it has been involved in, highlighting the potential suspicious activity it has detected in the same colour as the dot. For instance, if a vessel is predicted to have been involved in drug trafficking, their dot turns red and the vessel information (name, type, flag) appears with a list of their activities with and the specific *drug trafficking* activity and the information associated with it (time, location, date, and certainty level) highlighted in red. If a vessel is predicted to be involved in drug trafficking and illegal fishing, the dot turns orange and the list will highlight the activity and associated information for *drug trafficking* in red and *illegal fishing* in purple. It is also important to note that there should be a legend to label what activities the colours mean, or provide instructions that are visible on the interface so users understand how to interpret the colours.

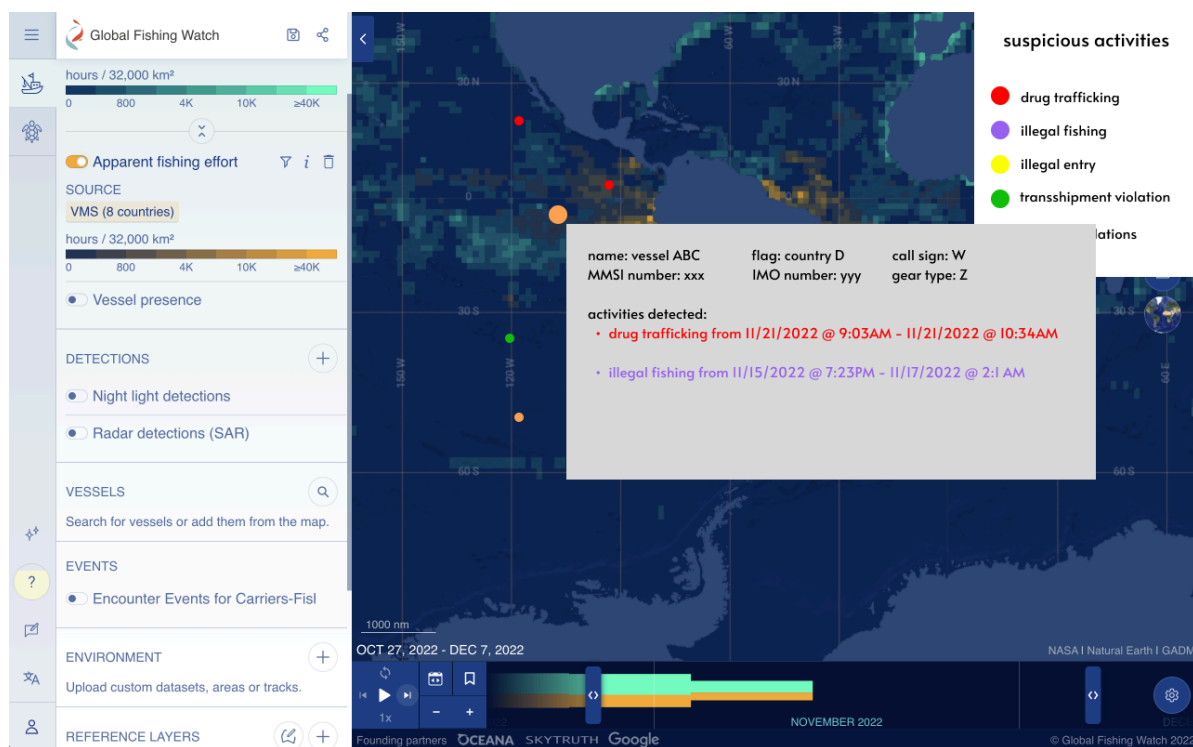


Figure 3: Colour code activities prototype with an orange dot clicked

New users may not be aware that *Grace* can detect suspicious activities, so these coloured dots can signify such affordance. And, when users click on these dots, the pop-up modal displays vessel information and the activities they are involved in, providing good feedback.

Search Bar

Another way to implement an **affordance** is to ensure *Grace* provides the functionalities a user would expect it to have. From our observations, we saw many users misinterpreting the vessel search bar functionality, typing in abbreviations for countries and expecting the system to return vessels of that flag. Instead, GFW returned vessels with those letters in their name. For better user experience, *Grace* should offer the ability to search for a specific flagged vessel, the vessel name, or other properties since this is what the users expect. This fulfills the functional requirement of detecting and monitoring vessels as it provides features to help users carry out this task. For the **signifier**, we can take advantage of users' transfer knowledge from other types of interfaces, as this makes learning and remembering how to use the interface faster and more easily. Then, the **feedback** here should match their expectations, meaning any time they perform an action, the correct and appropriate results should appear.

In figure 4, we designed a search bar. A search bar is a good cue to let users know they can search for specific vessels since it is used across different types of platforms to indicate the action to search. However, it is only a good cue if it is visible, thus, we recommend *Grace* to have the search bar stick to the top of the interface so that it is visible at point of entry but also when users move through the interface. Besides visibility, it matches where search functions are on other types of interfaces like search engines, shopping websites, streaming platforms, and more, making it a very common location and one of the first places users would look. And, from a cultural perspective, most cultures read top to bottom, making the search function easy to find if located at the top. Also, since we recommended *Grace* to be able to search for vessels by specific properties, we should implement signifiers, like buttons, to communicate this function to the user. The button selected by the user can be shaded a darker colour and the label inside the search bar can reflect this change, reading *search by X*, based on what has been selected. This notifies the user something has changed within the system. Then, it should return the appropriate list of vessels based on the search function and user input.

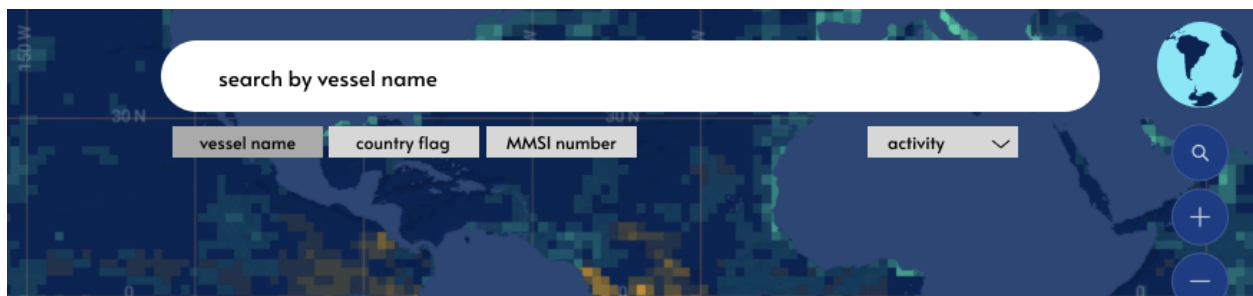


Figure 4. *Search bar prototype with vessel name property selected*

Menu Bar

Finally, we designed a menu bar to show how affordances, signifiers, and feedback can improve user experience (Figure 5 and 6). Since one of the **affordances** is to filter the system based on user criteria, we want to ensure users are aware they can use the *Filter* function. However, from the observations, we saw users were having difficulty locating the *Filter* feature, suggesting the icon independently was difficult to identify. Thus, we recommend using both an icon and text to **signify** the possible actions. This is because we understand cultural and individual differences exist, resulting in different capabilities, fallibilities, experiences, and values, which may hinder users in understanding these signifiers correctly. We also understand that not every image is universally accepted or understood, so one way to mitigate this issue is to complement the icon with text. Ideally, we could offer different languages for the interface as well which makes it even more accessible. Then, by providing both an image and label, users have two sources to depend on when looking for specific features. Furthermore, we recommend using dropdowns to encompass the features as it provides a cleaner and less overwhelming look and feel to the interface. Dropdowns also minimizes human error since users can only select a number of dropdowns until they find the feature they are looking for. We also used checkmark boxes in our design as it allows users to select and deselect the features they desire, providing immediate **feedback** as users can see a checkmark appearing and disappearing based on their actions. Moreover, to ensure users are always aware of the system status, we include numbers to indicate the number of filters they have applied.

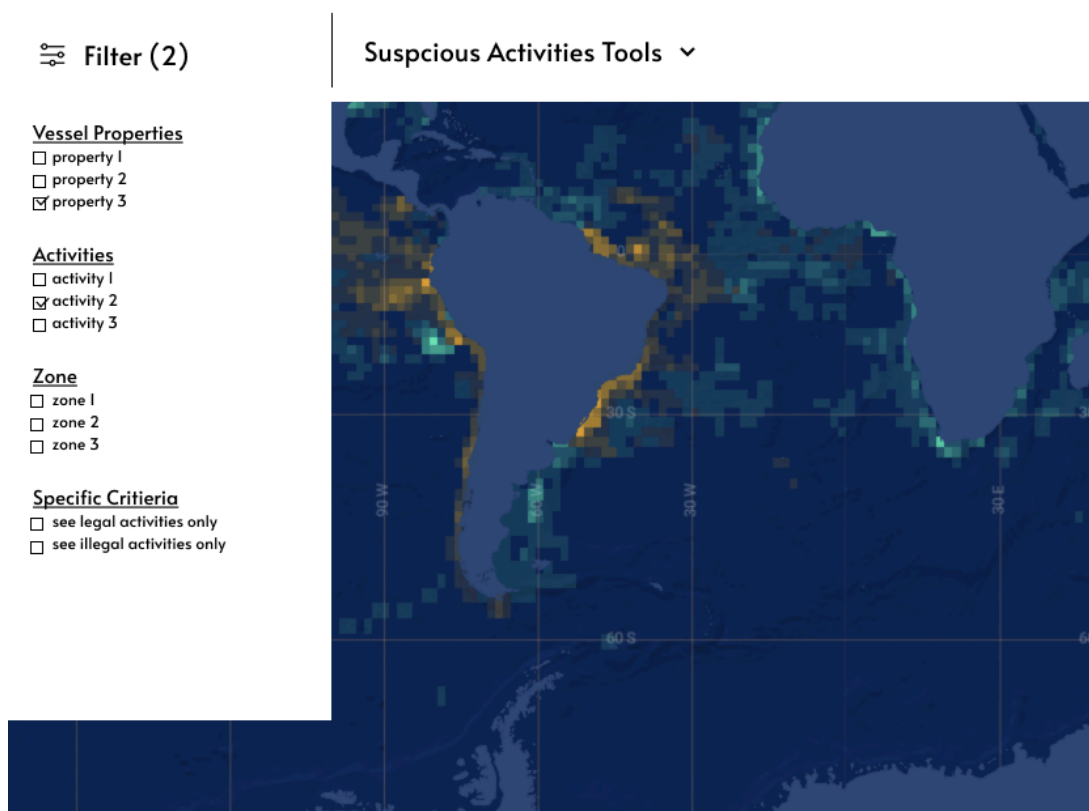


Figure 5: Menu bar prototype with filter dropdown opened

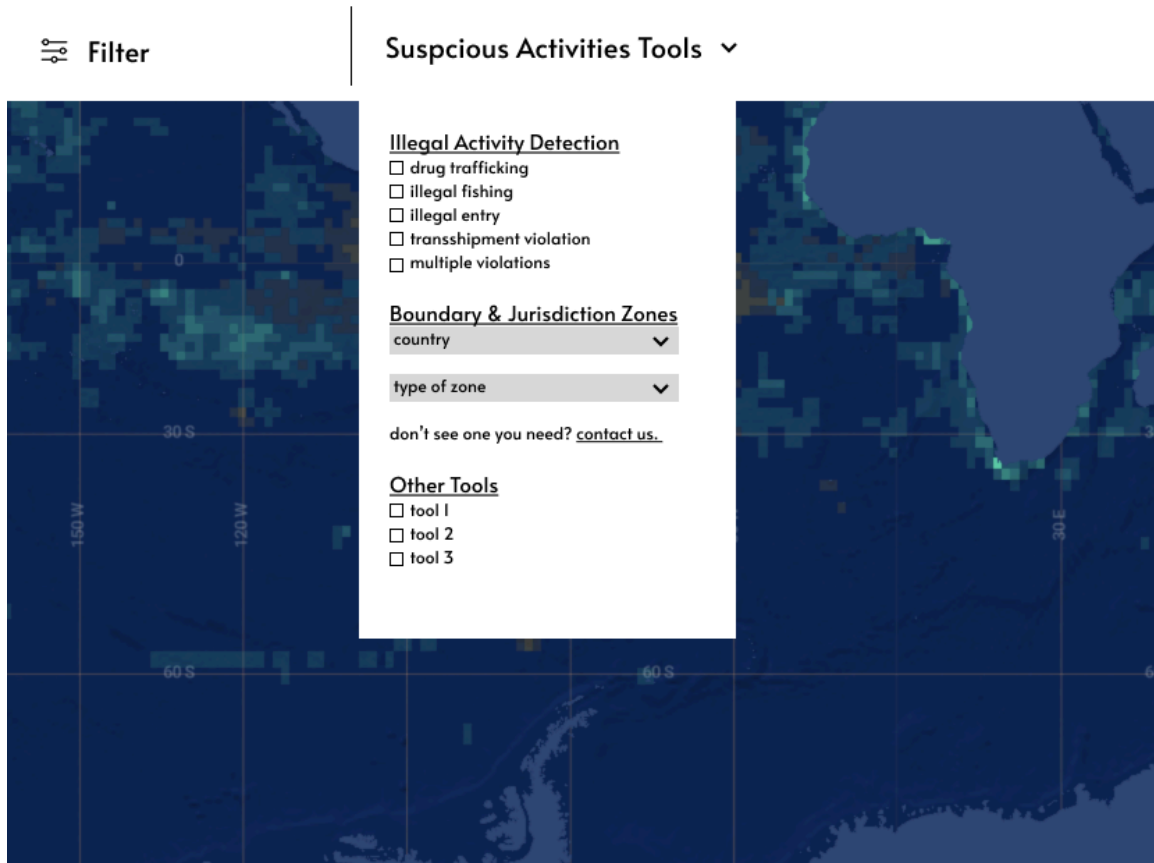


Figure 6: Menu bar prototype with suspicious activities tool dropdown opened

Summary

As *Grace* is an MDA platform offering a variety of features and information, it must be organized in a way so users understand how to approach and use the interface, and also what has occurred based on their actions. To help with this, we identified affordances, signifiers, and feedback to be the critical concepts to consider while designing the interface with regards to the requirements we identified from our research. Affordances help fulfill the functional requirement, signifiers help with learnability and retention, and the way we design signifiers and feedback allows features to be globally understood. By keeping these concepts in mind, we can improve usability and user experience, and minimize their errors.

Limitations

Due to time and resource constraints, we were unable to schedule more end users based internationally. It was very difficult scheduling times that worked for both the researcher and the participants located internationally due to the time difference and internet connection. For a future study that has more time,

we recommend prioritizing users based around Africa and the Middle East to compare their usability and behaviours on MDA platforms. Furthermore, for our pre-design stage, we did not include as many personal questions regarding their culture which would be helpful when recommending more specific design solutions in the early- and mid-design stage.

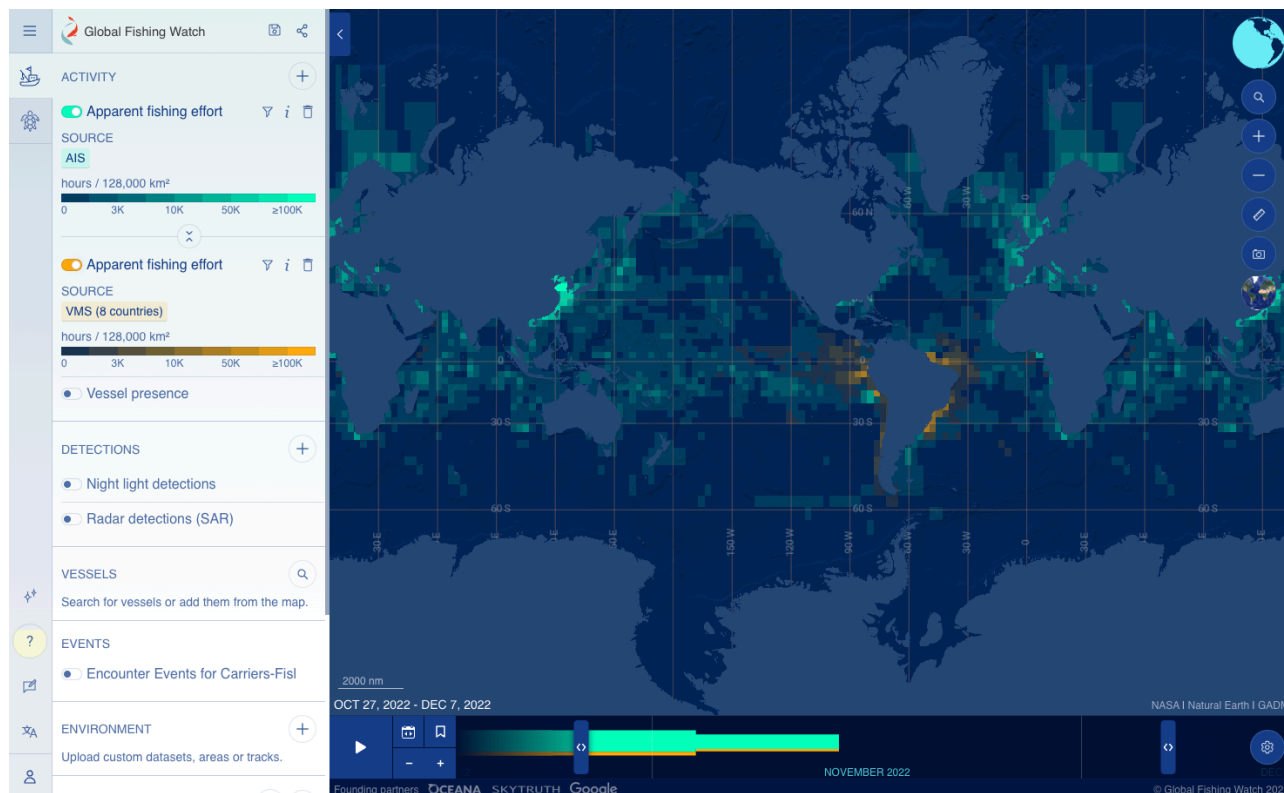
Another limitation was that SK required an account which made it more difficult to find users with an account already. However, because the internal stakeholders use SK and found the interface very helpful, this MDA platform was taken into consideration. However, for a future study with more resources, we would create our own account and let prospective users use these accounts for a better representation of the usability and user experience for SK.

Future Directions

Using this report, we recommend continuing the early design. This includes creating multiple lo-fi prototypes for *Grace* based on the requirements identified and our recommendations for design concepts. Although we created a few lo-fi prototypes for specific features, they are “throw-away” prototypes with the purpose of generating ideas. We recommend designers to conduct cognitive walkthroughs to understand what works and what does not, reiterating on the design until satisfied. If time allows, we also recommend A/B testing with end users to understand which of the lo-fi prototypes work best and why, before entering into the mid-design stage where designs are refined to become more med-fi. The med-fi prototypes should be tested using usability tests and heuristic evaluations by experts within the maritime field, then using the feedback to refine their design into a high-fi prototype. Finally, in the late-design stage, the prototype should be functional and in the beta phase, ready to be deployed. All the specifications should be completed and this hi-fi prototype should be ready for field testing and evaluation by real end users.

APPENDIX A

A.1) Global Fishing Watch Interface



A.2) Global Fishing Watch Pop-Up Modal of Vessel Information

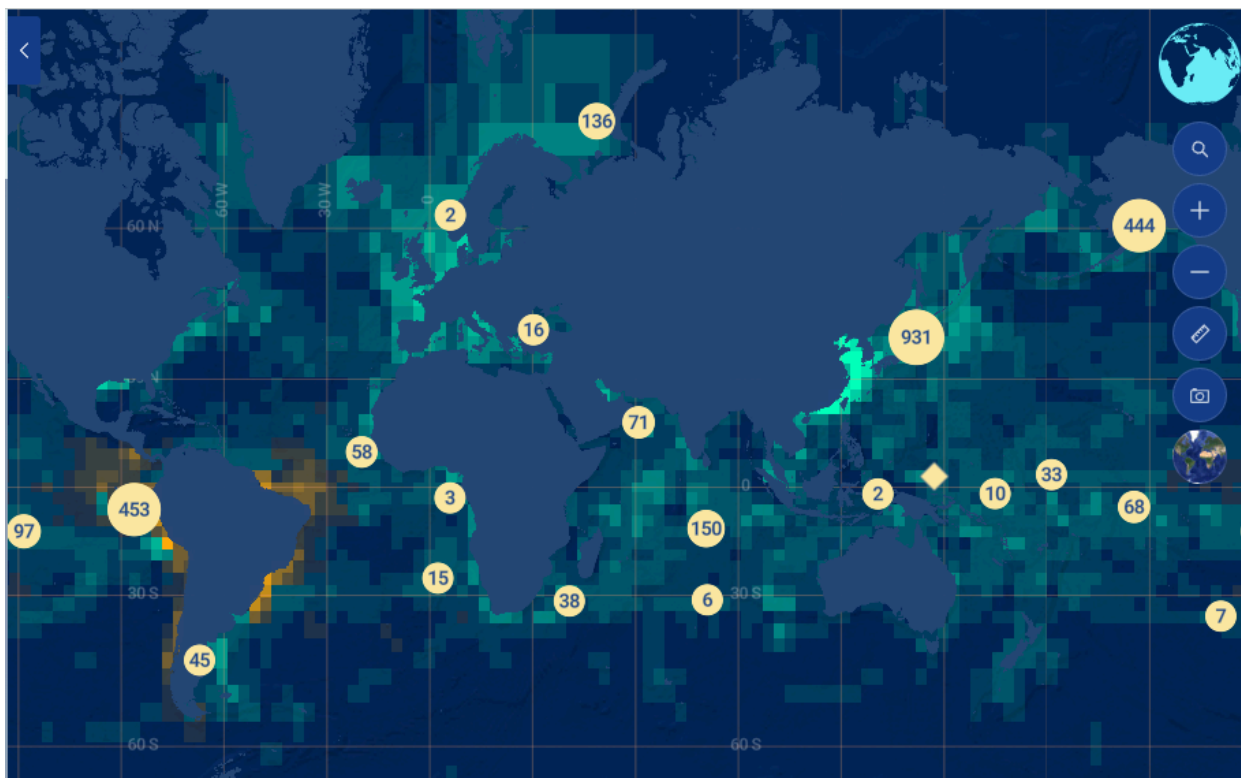
Apparent fishing effort between Aug 6, 2022 and Nov 4, 2022

4,198 hours

VESSELS	FLAG	GEAR	SOURCE	HOURS
TAIXIANG1	CHN	Drifting longline	AIS (Fishing Ves...	277
Tai Xiang 10	CHN	Drifting longline	AIS (Fishing Ves...	261
Tai Hong 6	CHN	Drifting longline	AIS (Fishing Ves...	259
Imula 0013 Jfn	LKA	Drifting longline	AIS (Fishing Ves...	245
Imula 0682 Chw	LKA	Drifting longline	AIS (Fishing Ves...	225

+ 45 more

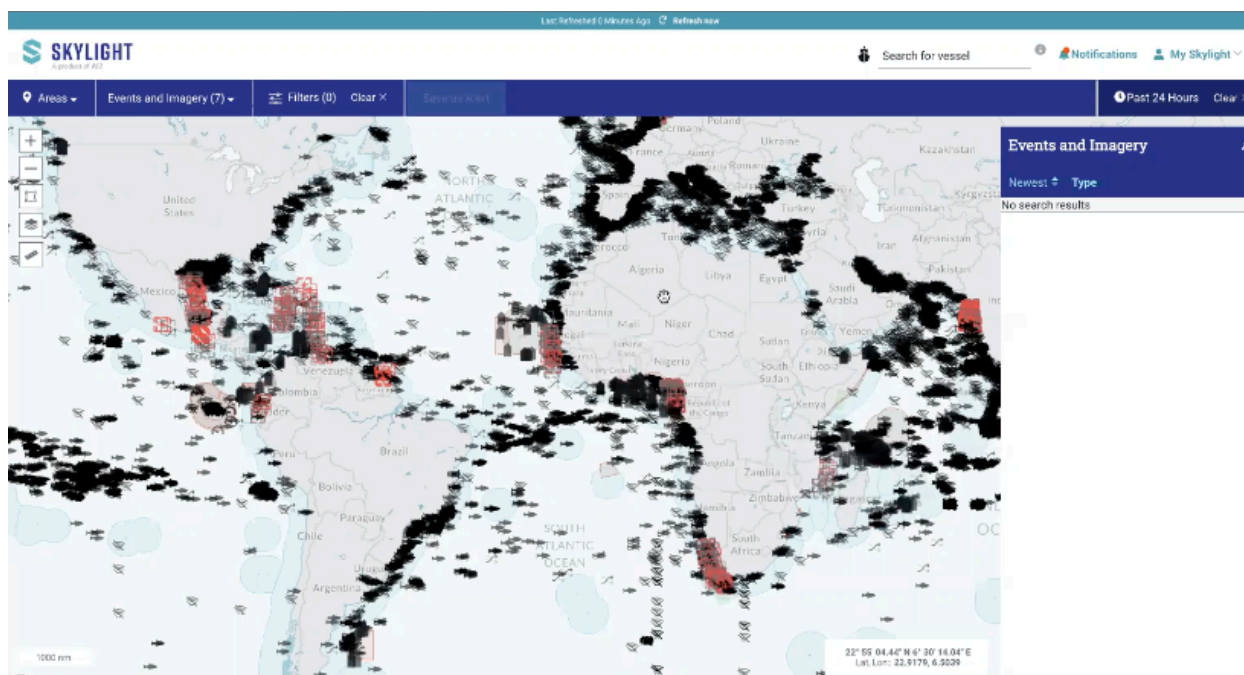
A.3) Global Fishing Watch Encounters Result



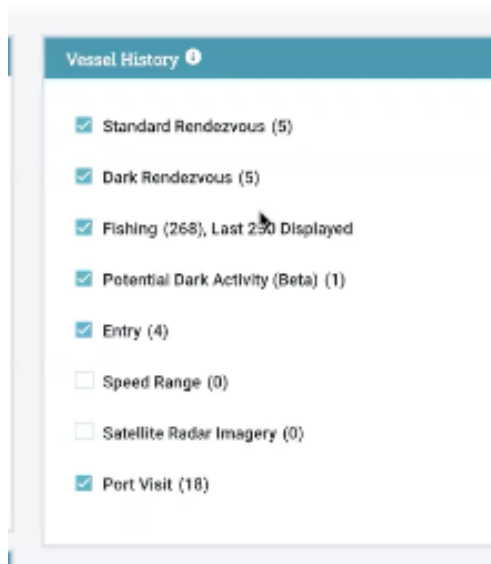
A.4) Question Mark Bubble



B.1) Skylight Interface



B.2) Skylight Events & Imagery Dropdown



C.1) GFW Affinity Diagram

D.3) Competitive Review Raw Data

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